

From Observable Causal Order to Conformal Cone Geometry

Operational Scale Selection and Cross-Stratum Validation in Resolution Geometry

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August 20, 2026

Abstract

We formulate an observation cone without deriving it from astronomical data or postulating Einstein dynamics. Begin with the universal observational quotient Q_{obs} and equip it with a fibre-compatible observational causal preorder. Its future sets $J_{\text{obs}}^+(x)$ and their boundaries define the observable causal cone. When this ordered set admits a time-oriented Lorentzian manifold realization of dimension greater than two satisfying future and past distinguishability, the Hawking–King–McCarthy–Malament reconstruction results imply that the causal order determines the topology, differentiable structure, and Lorentzian metric up to positive conformal rescaling. Thus the mathematically justified bridge is conditional:

$$(Q_{\text{obs}}, \preceq_{\text{obs}}) + \text{Lorentzian realizability and causality} \implies [g_{\text{obs}}],$$

not an unconditional derivation of spacetime from an arbitrary preorder.

The remaining conformal factor requires scale data. On a multiplicative, zero-memory coherence stratum, $\mathcal{T} = -\log C$ is additive and can serve as an operational duration. We prove, however, that one calibrated worldline does not determine a spacetime-wide conformal factor. Metric selection requires a sufficiently local clock field, an equivalent volume element, or other scale calibration. Astronomy is then assigned its correct role: a remote cross-stratum test of the calibrated null cone. A synthetic redshift–distance law identifies only a ratio such as v_{∂}/H_0 until independent calibration is supplied. The final deterministic attack, seed 20260820, used 150000 samples and passed all 14 registered checks, including conformal invariance, quotient descent, memory violation, scale underdetermination, astronomical degeneracy, and redshift-dependent falsification. The result characterizes causal/conformal observation geometry without claiming the Einstein equations or identifying the cone scale by mathematics alone.

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1 Claim boundary

The objective is to isolate what follows from observation, what follows only after a Lorentzian realization hypothesis, and what remains empirical physics. The paper establishes the ladder

$$\begin{aligned} \text{observable quotient} &\rightarrow \text{observable causal order} \rightarrow \text{conditional conformal realization} \rightarrow \\ &\rightarrow \text{scale calibration} \rightarrow \text{cross-stratum test}. \end{aligned}$$

It does not derive causal order from finite bandwidth, does not derive a smooth manifold from an arbitrary set, and does not derive gravitational dynamics.

2 Observation quotient and causal descent

Let X_{lat} be a class of latent realizations and $\mathfrak{F} : X_{\text{lat}} \rightarrow X_{\text{obs}}$ the declared observation law. Define $x \sim_{\text{obs}} x'$ iff $\mathfrak{F}(x) = \mathfrak{F}(x')$, and let $\pi : X_{\text{lat}} \rightarrow Q_{\text{obs}} = X_{\text{lat}} / \sim_{\text{obs}}$ be the quotient map.

Definition 2.1 (Latent observational reachability). A relation $\preceq_L$ on X_{lat} is an observational reachability preorder when it is reflexive and transitive and when $x \preceq_L y$ means that, under the declared admissible intervention family, an influence initiated at x is observationally distinguishable as capable of reaching y .

Definition 2.2 (Fibre compatibility). The relation $\preceq_L$ is fibre compatible when

$$x \sim_{\text{obs}} x', \quad y \sim_{\text{obs}} y' \implies (x \preceq_L y \iff x' \preceq_L y').$$

Theorem 2.3 (Observational causal quotient). A latent reachability relation descends by

$$[x] \preceq_{\text{obs}} [y] \iff x \preceq_L y$$

to a well-defined preorder on Q_{obs} if it is fibre compatible. Conversely, well-defined descent implies fibre compatibility.

Proof. Fibre compatibility makes the definition independent of representatives; reflexivity and transitivity descend immediately. Conversely, if the quotient relation is well defined, replacing either representative cannot alter its truth value, which is fibre compatibility. $\square$

Remark 2.4. The quotient theorem alone does not manufacture causal order. Reachability is additional observational structure that must be specified or recovered from an admissible intervention family. Theorem 2.3 states the exact gate required for that structure to live on the observational quotient.

Definition 2.5 (Observation cone). For $q \in Q_{\text{obs}}$, define

$$J_{\text{obs}}^+(q) = \{q' \in Q_{\text{obs}} : q \preceq_{\text{obs}} q'\}.$$

Whenever Q_{obs} carries a topology compatible with the realization below, the future observation cone is the boundary

$$\mathcal{C}_{\text{obs}}^+(q) = \partial J_{\text{obs}}^+(q).$$

3 From causal order to conformal geometry

An arbitrary preorder need not be a Lorentzian spacetime. The established reconstruction theorem applies only after explicit realizability, dimensional, and causality assumptions.

Assumption 3.1 (Lorentzian realization gate). The ordered observational quotient $(Q_{\text{obs}}, \preceq_{\text{obs}})$ is isomorphic to the causal order of a connected, time-oriented Lorentzian manifold (M, g) of dimension $n > 2$. The realization is future and past distinguishing. The order is understood together with the time orientation required to distinguish future from past.

Theorem 3.2 (Conditional causal reconstruction). *Under Assumption 3.1, the observational causal order determines the Lorentzian conformal geometry: any order isomorphism to another distinguishing Lorentzian realization is a conformal diffeomorphism. Hence the observation structure determines a conformal class*

$$[g_{\text{obs}}] = \{\Omega^2 g : \Omega > 0\}.$$

Proof. This is the Hawking–King–McCarthy–Malament reconstruction result specialized to the observational realization: in the relevant distinguishing causality class, causal structure determines the manifold/conformal structure up to conformal isometry [1, 2, 3]. $\square$

Corollary 3.3 (Conformal invariance of the observation cone). *Positive rescaling $g \mapsto \Omega^2 g$ preserves causal vectors, causal order, and null boundaries. Therefore $\mathcal{C}_{\text{obs}}^+$ is conformal data and does not determine absolute metric scale.*

Proof. For every tangent vector v , $(\Omega^2 g)(v, v) = \Omega^2 g(v, v)$ has the same sign because $\Omega > 0$. Time orientation fixed, causal futures and their null boundaries are unchanged. $\square$

Non-claim 3.4. Theorem 3.2 is an established Lorentzian reconstruction theorem, not a new theorem of Resolution Geometry. The contribution here is the typed assembly: observational quotient, fibre-compatible reachability, explicit realization gate, and separation of conformal reconstruction from scale and dynamics.

4 Operational duration and the scale lock

Let $C_{ij} \in (0, 1]$ be a measured coherence assigned to a composable segment. Define

$$\mathcal{T}_{ij} = -\log C_{ij}.$$

Proposition 4.1 (Additive zero-memory clock). *If coherence composition is multiplicative, $C_{02} = C_{01}C_{12}$, then*

$$\mathcal{T}_{02} = \mathcal{T}_{01} + \mathcal{T}_{12}.$$

If instead

$$\mathcal{T}_{02} = \mathcal{T}_{01} + \mathcal{T}_{12} + \mathcal{M}_{012},$$

strict additivity holds exactly on the stratum $\mathcal{M}_{012} = 0$.

Proof. Apply $-\log$ to multiplication. The second statement follows by subtracting the two additive terms. $\square$

Additivity creates an operational duration, but this alone does not globally select a representative of $[g]$.

Theorem 4.2 (Single-worldline scale underdetermination). *Calibration of proper duration along one timelike curve does not, in general, determine the conformal factor on the surrounding spacetime.*

Proof. In coordinates (t, x) near the curve $x = 0$, take $g_1 = g$ and $g_2 = \Omega^2 g$ with

$$\Omega(t, x) = 1 + \frac{x^2}{4}.$$

The two metrics agree along $x = 0$ and therefore generate the same calibrated clock there, but differ for $x \neq 0$. Both belong to the same conformal class. $\square$

Corollary 4.3 (Sufficient scale data). *A representative metric can be selected when the operational calibration supplies the conformal factor locally throughout the region of interest, or when equivalent local scale data such as a compatible volume element are given. The selected metric is unique within the conformal class wherever that local scale datum is fixed.*

Proof. If $g' = \Omega^2 g$, pointwise scale data determine the positive scalar Ω . A volume form in dimension n transforms as $d\mu_{g'} = \Omega^n d\mu_g$ and therefore also fixes Ω pointwise. $\square$

Remark 4.4 (Precise role of the coherence clock). The clock is a candidate scale datum only after calibration and locality are established. Proposition 4.1 proves composition on the zero-memory stratum; it does not prove that $\mathcal{T}$ equals GR proper time or that one clock fixes an entire metric field.

5 Astronomy as cross-stratum validation

Astronomy is not used to generate the cone. After local reconstruction and scale calibration, remote null propagation tests whether the same cone structure is realized across observational strata.

Let a simplified redshift–distance family have the form

$$d(z) = \frac{v_\partial}{H_0} f(z; \vartheta),$$

where ϑ denotes other declared dimensionless parameters.

Proposition 5.1 (Astronomical scale degeneracy). *With f and ϑ fixed, the redshift–distance law identifies the ratio v_∂/H_0 , not v_∂ and H_0 separately. Independent calibration of either factor is required to identify the other.*

Proof. For every $a > 0$, replacing (v_∂, H_0) by (av_∂, aH_0) leaves $d(z)$ unchanged. $\square$

Definition 5.2 (Cross-stratum cone invariance test). For admissible strata α , compare their reconstructed conformal structures and, after independent scale calibration, their boundary speeds:

$$[g_\alpha] \stackrel{?}{=} [g_{\text{obs}}], \quad v_\partial^{(\alpha)} \stackrel{?}{=} v_\partial.$$

A redshift-dependent alternative $v_\partial(z)$ is tested against the constant boundary model by its structured residual across z .

Remark 5.3. The identification $v_\partial = c$ is empirical physics: calibrated optical and astronomical null propagation identify the reconstructed boundary with the experimentally established light cone. It is not a consequence of quotient universality or causal reconstruction alone.

6 Four-lock closure theorem

Theorem 6.1 (Observation-cone architecture). *Subject to the hypotheses stated in the preceding sections, the observation-cone construction separates into four locks:*

1. Quotient lock: *observational equivalence produces Q_{obs} , and fibre-compatible reachability descends to $\preceq_{\text{obs}}$.*
2. Conformal lock: *Lorentzian realizability plus distinguishing causality permits causal-order reconstruction of $[g_{\text{obs}}]$.*
3. Scale lock: *an additive operational clock is available on the zero-memory stratum, but a metric representative requires sufficiently local calibration or equivalent scale data.*
4. Cross-stratum lock: *astronomy tests remote conformal/null consistency and, after independent calibration, the universality of v_{∂} .*

None of these locks entails Einstein dynamics.

Proof. Item 1 is Theorem 2.3; item 2 is Theorem 3.2; item 3 follows from Proposition 4.1 and Theorem 4.2; item 4 is the operational test following Proposition 5.1. The Einstein equations require additional dynamical fields and laws absent from these premises. $\square$

7 Final deterministic attack

The proofs are analytic. The battery is a reproducibility and hostile-boundary check. The final run used seed 20260820, 150000 randomized samples, and passed all 14 registered checks.

Attack	Result	Meaning
Conformal cone classification	pass	Zero sign mismatches.
Conformal causal order	pass	Zero future-order mismatches.
Quotient descent	pass	Fibre-compatible relation is well defined.
Hostile fibre perturbation	pass	Noncompatible relation is rejected.
Log-coherence composition	pass	Residual 8.88×10^{-16} .
Memory attack	pass	Mean nonadditivity 0.15989.
Single-worldline scale attack	pass	Off-curve conformal gap 0.43686.
Local clock-field recovery	pass	Relative RMSE 1.005×10^{-4} .
Astronomical ratio degeneracy	pass	Exact matched law under common scaling.
Independent calibration	pass	Relative boundary-speed bias 8.25×10^{-5} .
Redshift-dependent violation	pass	Fractional structured residual 0.00785.
Cross-stratum consistency	pass	Synthetic consistency score 0.0482.

The attack sharpened rather than merely confirmed the proposal. In particular, it rejected the stronger claim that one operational clock automatically fixes a global conformal factor. The corrected scale theorem requires local clock-field or equivalent volume calibration.

8 Reproducibility record

The final artifacts are

- `observation_cone_final_attack.py`;
- `observation_cone_final_attack.json`.

The JSON records the seed, sample count, thresholds, all check outcomes, and the scope verdict. The astronomy component is synthetic and tests identifiability; it is not a fit to observational catalogues. All numerical checks are metadata and are not premises of Theorem 6.1.

9 Non-claims and conclusion

Non-claim 9.1. This paper does not prove that every observational preorder has a Lorentzian realization, does not derive smooth manifold structure from arbitrary records, does not prove that coherence duration equals proper time, does not derive the Einstein equations or a stress tensor, and does not estimate c or H_0 from real astronomical data.

The defensible result is both strong and bounded. Observation supplies a universal quotient. An additional fibre-compatible reachability structure can supply observational causal order. Under explicit Lorentzian realizability and causality gates, established reconstruction mathematics turns that order into a conformal cone geometry. Operational duration can select scale only when it is additive and sufficiently locally calibrated. Astronomy then tests whether the same calibrated null structure survives remotely; it does not create the cone or independently determine absolute scale.

Observable distinguishability plus admissible reachability yields an observational causal order.

A distinguishing Lorentzian realization turns that order into conformal cone geometry.

Local operational calibration selects metric scale.

Astronomy tests cross-stratum invariance.

Gravitational dynamics remains additional physics.

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